

3(a)	Hooke's (law)	B1
3(b)(i)	$k = F / x$ or $k = \text{gradient}$ e.g. $k = 7.0 / 5.0 \times 10^{-2}$	C1
	$= 140 \text{ N m}^{-1}$	A1
3(b)(ii)	$E = \frac{1}{2} F x$ or $E = \frac{1}{2} k x^2$ or $E = \text{area under graph}$ $= \frac{1}{2} \times 5.6 \times 4.0 \times 10^{-2}$ or $\frac{1}{2} \times 140 \times (4.0 \times 10^{-2})^2$	C1
	$= 0.11 \text{ J}$	A1
3(c)(i)	(upthrust =) $6.20 - 5.60 = 0.60 \text{ (N)}$	A1
3(c)(ii)	$\Delta p = \Delta F / A$ $= 0.60 / 1.2 \times 10^{-3}$	C1
	$= 500 \text{ Pa}$	A1
3(c)(iii)	$(\Delta)p = \rho g(\Delta)h$ $\rho = 500 / (9.81 \times 5.8 \times 10^{-2})$	C1
	$= 880 \text{ kg m}^{-3}$	A1
3(d)(i)	(upthrust) increases	B1
3(d)(ii)	(extension) decreases	B1